

Heat transport in chaotic magnetic fields

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1 PyOculus workflow

In `PyOculus`, we can define a magnetic configuration by specifying the components of the magnetic field $\mathbf{B}$. In our case, we want to study the following magnetic field in slab geometry, in which a perturbation creates an island structure:

$$\begin{cases} B_x = m\epsilon(x) \sin \alpha \\ B_y = \iota'(x - x_0) + \epsilon'(x) \cos \alpha \\ B_z = 1 \end{cases} \quad (1)$$

For a system with $x \in [0, 1]$ containing an island at $x_s = 0.5$, as a simple model for a perturbation localized around the rational surface, we choose

$$\epsilon(x) = \delta \cdot x(1 - x). \quad (2)$$

2 Modifications on PyOculus

In order to reproduce the magnetic configurations in slab geometry I worked with on TPIV2, I had to introduce some modifications on the `PyOculus` library.

2.1 Magnetic field in the slab model

I created two new classes:

- `SlabBfieldOnePert()`: Stemming from the `ToroidalBfield()` class, this class implements a magnetic field containing one magnetic island in slab geometry.
- `SlabBfieldTwoPerts()`: Stemming from the `ToroidalBfield()` class, this class implements a magnetic field containing two magnetic islands in slab geometry.

These changes haven't been merged into the package, for now they are implemented in files located in the `SdS/local/pyoculus_mod` folder.

2.2 Change of the domain of the toroidal map

In order to generate the map corresponding to a magnetic field in slab geometry, I use the class `ToroidalBfieldSection()`. This class had the domain of the x coordinate set to $[-1, 1]$. Since in some cases I want to work with a different domain, I introduced a small modification in the definition of the class, so that a user can specify the domain of x when creating an instance of the class, while preserving $[-1, 1]$ as the default domain.

This change has been made in the forked version of the library, in the file `pyoculus/pyoculus/maps/toroidal` lines 18-19, and committed and pushed in the branch `slab-model`.